

Linear Transformation

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Chapter 5

Linear Transformation and Transformation Matrix

1 Linear Transformation and Transformation Matrix

2 Kernel and Image



Transformation

Definition

A transformation $T : V \rightarrow W$ assigns to each $x \in V$ a vector $T(x) \in W$.

- V : the domain of T
- W : the codomain of T
- $T(x)$: the image of x
- Image/range of $T : \{T(x) : x \in V\}$
- The preimage of $y : x$ such that $T(x) = y$



Operations on Transformations

- ① Addition: $(T_1 + T_2)(x) = T_1(x) + T_2(x)$
- ② Scalar multiplication: $(\lambda T)(x) = \lambda T(x)$
- ③ Composition: $(T_1 T_2)(x) = T_1(T_2(x))$
- ④ Inverse: $T^{-1}(T(x)) = x$



Linear Transformation

Definition

The transformation $T : V \rightarrow W$ is linear if

① $T(v_1 + v_2) = T(v_1) + T(v_2),$

② $T(\lambda v) = \lambda T(v).$

for any $v_1, v_2 \in V$ and any scalar λ

If $W = V$, then T is also called a linear transformation of V .



Lemma

Let $T : V \rightarrow W$ be a linear transformation and $E = \{e_1, \dots, e_n\}$ be a basis of V . Let $x \in V$. If $[x]_E = (x_1, x_2, \dots, x_n)$, then

$$T(x) = x_1 T(e_1) + x_2 T(e_2) + \dots + x_n T(e_n).$$



Transformation Matrix

Let $E = \{e_1, \dots, e_n\}$ and $F = \{f_1, \dots, f_m\}$ are bases of V and W , respectively. If $[T(x)]_F = A_{FE}[x]_E$ for all x then

$$[T(e_i)]_F = A_{FE}[e_i]_E = A_{FE}[0 \dots 1 \dots 0]^T = \text{the } i\text{-th column of } A_{FE}.$$

then

$$A_{FE} = [[T(e_1)]_F \cdots [T(e_n)]_F].$$



Transformation Matrix

Definition

Let $T : V \rightarrow W$ be a linear transformation. Assume E and F are bases of V and W , respectively. Then

$$A_{FE} = \left[[T(e_1)]_F \cdots [T(e_n)]_F \right].$$

is called the **transformation matrix** or the **matrix representation** of T w.r.t E and F . Moreover,

$$[T(x)]_F = A_{FE}[x]_E.$$

- If $F = E$, then we denote $A_E = A_{EE}$.
- If E and F are standard bases, then A is called the **standard matrix**.

Theorem

Assume E and F are respectively bases of finite dimensional vector spaces V and W .

- a** For any linear transformation $T : V \rightarrow W$, there is a unique matrix A such that

$$[T(x)]_F = A[x]_E.$$

- b** For any matrix A over K , there is a unique linear transformation $T : K^n \rightarrow K^m$ such that $[T(x)]_F = A[x]_E$.



Properties of transformation matrices

- ① $[T_1 + \alpha T_2]_{FE} = [T_1]_{FE} + \alpha[T_2]_{FE}$
- ② $[T_2 T_1]_{GE} = [T_2]_{GF} \cdot [T_1]_{FE}$
- ③ $[T^n]_E = [T]_E^n$
- ④ $[T^{-1}]_{EF} = [T]_{FE}^{-1}$.



Kernel and Image

1 Linear Transformation and Transformation Matrix

2 **Kernel and Image**

